

Cauchy-Riemann equation.

Let $f: G \rightarrow \mathbb{C}$ be a map on region G and let $U(x,y) = \operatorname{Re} f(x,y)$, $V(x,y) = \operatorname{Im} f(x,y)$. Suppose that U and V have continuous partial derivatives. Then,

f is analytic if and only if U and V satisfy the Cauchy-Riemann equation.

Proof Suppose that f is analytic. For $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+h) - f(z)}{h} = \frac{U(x+h,y) - U(x,y)}{h} + i \cdot \frac{V(x+h,y) - V(x,y)}{h}$$

Letting $h \rightarrow 0$, then $f'(z) = \frac{\partial U}{\partial x}(x,y) + i \cdot \frac{\partial V}{\partial x}(x,y)$.

Meanwhile, for $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+ih) - f(z)}{i \cdot h} = -i \cdot \frac{U(x,y+h) - U(x,y)}{h} + i \cdot \frac{V(x,y+h) - V(x,y)}{h}$$

Letting $h \rightarrow 0$, then $f'(z) = -i \cdot \frac{\partial U}{\partial y}(x,y) + \frac{\partial V}{\partial y}(x,y)$

So, $U_x = V_y$ and $V_x = -U_y$. (*) [not using continuously differentiable.]

Conversely, U and V satisfy the condition $(*)$, and u_x, u_y, v_x, v_y are continuous. Let $z = x + iy \in G$ be given. Since G is open, for some $r > 0$, $B_r(z) \subseteq G$. For $h = s + it \in B_r(0)$,

$$U(x+s, y+t) - U(x, y) = [U(x+s, y+t) - U(x, y+t)] + [U(x, y+t) - U(x, y)] \quad (*)$$

By the Mean-Value Theorem for variables s and t , respectively,

$$U(x+s, y+t) - U(x, y+t) = U_x(x+s_1, y+t) \cdot s \quad \text{for some } |s_1| < |s|$$

and

$$U(x, y+t) - U(x, y) = U_y(x, y+t_1) \cdot t \quad \text{for some } |t_1| < |t|$$

Let $\varphi(s, t) = [U(x+s, y+t) - U(x, y)] - [U_x(x, y) \cdot s + U_y(x, y) \cdot t]$. Now, $(*)$ gives

$$\frac{\varphi(s, t)}{s + it} = \frac{s}{s + it} \cdot [U_x(x+s_1, y+t) - U_x(x, y)] + \frac{t}{s + it} \cdot [U_y(x, y+t_1) - U_y(x, y)]$$

But, $|s_1| < |s| \leq |s + it|$ and $|t_1| < |t| \leq |s + it|$, so letting $s + it \rightarrow 0$, then

$$\frac{\varphi(s, t)}{s + it} \rightarrow 0$$

Therefore, $U(x+s, y+t) - U(x, y) = U_x(x, y)s + U_y(x, y)t + \varphi(s, t)$

and similarly $V(x+s, y+t) - V(x, y) = V_x(x, y)s + V_y(x, y)t + \psi(s, t)$ where $\lim_{s+it \rightarrow 0} \frac{\psi(s, t)}{s+it} = 0$.

$$\text{Now, } \frac{f(z+s+it) - f(z)}{s+it} = U_x(z) + iV_x(z) + \frac{\varphi(s, t) + i\psi(s, t)}{s+it} \rightarrow U_x(z) + iV_x(z).$$

as $s + it \rightarrow 0$.

Lemma. Let $f: G \rightarrow \mathbb{C}$ be a holomorphic at $z_0 \in G$ where G is an open set. Then, for the differential operators

$$\frac{\partial}{\partial z} \stackrel{\text{def}}{=} \frac{1}{2} \cdot \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) \quad \text{and} \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \cdot \left(\frac{\partial}{\partial x} - \frac{1}{i} \cdot \frac{\partial}{\partial y} \right),$$

$$\frac{\partial f}{\partial \bar{z}}(z_0) = 0 \quad \text{and} \quad f'(z_0) = \frac{\partial f}{\partial z}(z_0) = 2 \cdot \frac{\partial u}{\partial z}(z_0),$$

Proof. Denote $f(x+iy) = u(x+iy) + i \cdot v(x+iy)$, then

$$\begin{aligned} \frac{\partial f}{\partial \bar{z}} &= \frac{1}{2} \cdot \left(\frac{\partial u}{\partial x} - \frac{1}{i} \frac{\partial u}{\partial y} \right) + \frac{1}{2} \cdot \left(i \cdot \frac{\partial v}{\partial x} - \frac{\partial v}{\partial y} \right) \\ &= \frac{1}{2} \cdot \left[\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} + i \cdot \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \right] \end{aligned}$$

so, f satisfies the Cauchy-Riemann condition iff $\frac{\partial f}{\partial \bar{z}} = 0$.

$$\begin{aligned} \text{And, } \frac{\partial f}{\partial z} &= \frac{1}{2} \cdot \left(\frac{\partial u}{\partial x} + \frac{1}{i} \frac{\partial u}{\partial y} \right) + \frac{1}{2} \cdot \left(i \cdot \frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} \right) \\ &= \frac{1}{2} \cdot \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \frac{i}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{aligned}$$